



## MA 575 : Complex Analysis

Course Code : MA 575

Course Name : Complex Analysis

L-P-T-C: 3-1-0-4

Intended for: M.Sc./M.Tech./PhD/B.Tech

Prerequisites: None

Mutual Exclusion: None

Approval : 57th BoA

### Course Contents

- **Module 1:** Complex numbers and the point at infinity, Polar representation, logarithmic and trigonometric functions, Analytic functions, Cauchy-Riemann conditions, Power series, Harmonic conjugates, Mobius transformations, Mappings by elementary functions, Liouville's theorem and its applications, Conformal mappings. (14 Hours)
- **Module 2:** Index of a closed curve, Cauchy's theorem, Cauchy integral formula, Power series representation of analytic functions, Open mapping theorem, Goursat's theorem. Uniform convergence of sequences and series. Taylor and Laurent series. (12 Hours)
- **Module 3:** Isolated singularities and residues, Residue theorem and its application to evaluation of real integrals. Zeroes and poles, Maximum Modulus Principle and Schwarz's lemma, Meromorphic functions, Argument Principle, Rouché's theorem, (16 Hours)

### Laboratory:

1. NA

### Text books:

1. J.B. Conway, **Functions of One Complex Variable**, 2nd Edition, Narosa, New Delhi, 1978.
2. T.W. Gamelin, **Complex Analysis**, Springer International Edition, 2001.
3. J.W. Brown and R.V. Churchill, **Complex variables and applications**, 9th edition, McGraw Hill Higher Education.

### References:

1. R. Remmert, **Theory of Complex Functions**, Springer Verlag, 1991.
2. A.R. Shastri, **An Introduction to Complex Analysis**, Macmillan India, New Delhi, 1999.
3. E.M. Stein and R. Shakarchi, **Complex analysis**, Princeton lecture series in analysis.
4. M.Thamban Nair, **Complex analysis online notes**, <https://home.iitm.ac.in/mtnair/ca.pdf>